

Benchmark C: Interoperable, cohort-scale targeted reanalysis

Recovering published biomarkers across a public cohort with *chromExtract()*

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Supplementary Material for:

Chromatograms: A modular infrastructure for scalable chromatographic data handling in R

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Motivation

Once a study has annotated a set of compounds (m/z and retention time), those annotations can be revisited in the raw data: pulled out and re-quantified, across the whole cohort or in new samples. Doing this the untargeted way - re-running peak detection, alignment and correspondence - is expensive. The [Metabonaut](#) large-scale analysis tutorial performs exactly that untargeted *xcms* workflow on the cohort used here, and it takes many hours of processing and hundreds of GB of intermediate data.

`Chromatograms::chromExtract()` offers a lightweight, targeted alternative. Given a plain peak table of m/z / retention-time windows, it pulls the corresponding extracted-ion chromatograms (EICs) directly from the raw data - no peak detection, no alignment - and returns a `Chromatograms` object. The recovered EICs are then quantified and quality-assessed with the package's own functions (`peakBoundary()`) together with *MsQuality*, and visualised with `plotChromatogramsOverlay()`. Because the workflow consumes *published annotations* and *raw public data*, it is tool-agnostic: the peak table can come from any study or software (*MZmine*, *MS-DIAL*, *xcms*, a vendor export, or, as here, a published table), and the resulting `Chromatograms` object flows on into any downstream analysis.

This document reanalyses **MTBLS93** (Ganna *et al.* 2014, *PLOS Genetics*, doi:10.1371/journal.pgen.1004801; the TwinGene cohort, ~4063 MS1 samples), recovering the coronary-heart-disease lipid biomarkers the study annotated. It demonstrates, using native `Chromatograms` functionality throughout:

1. targeted EIC extraction of the published biomarkers with `chromExtract()`;
2. quantification and quality assessment with `peakBoundary()` and *MsQuality*;
3. locating an annotation with no published retention time (SM 28:1) directly from the data;
4. how targeted extraction scales across the cohort (number of files).

```
library(Chromatograms)
library(Spectra)
library(MsBackendMetaboLights)
library(MsQuality)
library(BiocParallel)
library(peakRAM)

## Run serially so scaling reflects design, not the number of workers.
register(SerialParam())

## TEST RUN: small file count so the document renders quickly. The full-cohort
## run (hundreds/thousands of files) is done separately.
N_FILES <- 3L
RT_RANGE <- c(20, 900) # elution window (s)
```

System Information

```
knitr::kable(
  data.frame(
    Item = c("CPU Cores", "R Version", "Platform", "Chromatograms Version"),
    Value = c(
      parallel::detectCores(),
      R.version$version.string,
      R.version$platform,
      as.character(packageVersion("Chromatograms"))
    )
  ),
  caption = "System information"
)
```

Table 1: System information

Item	Value
CPU Cores	20
R Version	R version 4.6.0 (2026-04-24 ucrt)
Platform	x86_64-w64-mingw32
Chromatograms Version	1.3.2

1. Target compounds

The four replicated biomarkers from the study (MSI level 1). Both the m/z ($[M+H]^+$) and the retention times are taken from the article’s targeted MS/MS confirmation figure (S2 Figure), which reports elution times in minutes: LysoPC 18:1 at 6.37 min (382.2 s), LysoPC 18:2 at 5.78 min (346.8 s) and MG 18:2 at 6.40 min (384.0 s). SM 28:1 is confirmed by accurate mass (m/z 661.526) but the article gives **no** retention time for it; Section 2.1 shows how to locate it directly from the data before it joins the other targets.

```
targets <- data.frame(
  name = c("LysoPC_18:1", "LysoPC_18:2", "MG_18:2", "SM_28:1"),
  mz   = c(522.356, 520.341, 337.269, 661.526),
  rt   = c(382.2, 346.8, 384.0, NA), # S2 Figure (min x 60), SM 28:1 not given
  stringsAsFactors = FALSE
)
targets$mzmin <- targets$mz - 0.01
targets$mzmax <- targets$mz + 0.01
targets$rtmin <- targets$rt - 20
targets$rtmax <- targets$rt + 20

knitr::kable(targets, digits = 3, caption = "Annotated target biomarkers")
```

Table 2: Annotated target biomarkers

name	mz	rt	mzmin	mzmax	rtmin	rtmax
LysoPC_18:1	522.356	382.2	522.346	522.366	362.2	402.2
LysoPC_18:2	520.341	346.8	520.331	520.351	326.8	366.8
MG_18:2	337.269	384.0	337.259	337.279	364.0	404.0
SM_28:1	661.526	NA	661.516	661.536	NA	NA

2. Import and subset

MTBLS93 holds ~4063 MS1 files and ~800 GB of raw data, so we subset *before* touching the peaks: the file names are read from the assay table (metadata only), a deterministic subset (the first `N_FILES` files) is chosen, and only those files are loaded - `MsBackendMetaboLights` caches them on demand, so only the subset downloads. Retention time is restricted to the elution window to cut I/O.

```

assay <- mtbls_assay_data("MTBLS93")
ms1 <- unique(grep("01\\.CDF$",
                  assay[["Raw Spectral Data File"]], value = TRUE))
picked <- basename(head(ms1, N_FILES))
sps <- Spectra(source = MsBackendMetaboLights(), mtblsId = "MTBLS93",
              filePattern = paste0("(", paste(picked, collapse = "|"), ")$"))
sps <- filterRt(sps, rt = RT_RANGE)
files <- unique(dataOrigin(sps))

```

Table 3: Loaded MTBLS93 subset

Item	Value
Files	3
Total spectra	5949
RT range (s)	20.3 - 899.8

2.1 Locating an unannotated retention time (SM 28:1)

The article confirms SM 28:1 by accurate mass and MS/MS but does not report a retention time for it. Rather than borrow a value from an untargeted result, we locate it directly with the same targeted machinery, used here as a discovery step: extract the EIC across the *entire* elution window at the target m/z , see where the peak falls, then narrow the retention-time window and re-extract.

```

sm <- which(targets$name == "SM_28:1")

## SM 28:1 over the full elution window, one row per file
sm_table <- data.frame(
  name = "SM_28:1", mzMin = targets$mzmin[sm], mzMax = targets$mzmax[sm],
  rtMin = RT_RANGE[1], rtMax = RT_RANGE[2], msLevel = 1L, dataOrigin = files)
sm_chr <- filterEmptyChromatograms(
  chromExtract(Chromatograms(sps), peak.table = sm_table,
               by = c("msLevel", "dataOrigin")))

## consensus apex across the cohort: per-file apex retention time, median
apex_rt <- mapply(function(int, rt) rt[which.max(int)],
                  intensity(sm_chr), rtime(sm_chr))
sm_rt <- round(median(apex_rt), 2)

```

Over the full window a single peak stands out; its consensus apex across the files is at 579.85 s. We record that retention time and narrow the EIC we already extracted - `filterPeaksData()` trims each trace to the located window, no re-extraction from the raw data needed.

```

targets$rt[sm] <- sm_rt
targets$rtmin[sm] <- sm_rt - 20
targets$rtmax[sm] <- sm_rt + 20

sm_chr2 <- filterPeaksData(sm_chr, variables = "rtime",
                          ranges = c(sm_rt - 20, sm_rt + 20))

```

```

par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
Chromatograms::plotChromatogramsOverlay(
  sm_chr, col = "#0000AA55", main = "SM 28:1 - full window")
abline(v = sm_rt, col = "red", lty = 2)
Chromatograms::plotChromatogramsOverlay(
  sm_chr2, col = "#0000AA55",
  main = sprintf("SM 28:1 - narrowed (rt = %.1f s)", sm_rt))
abline(v = sm_rt, col = "red", lty = 2)

```

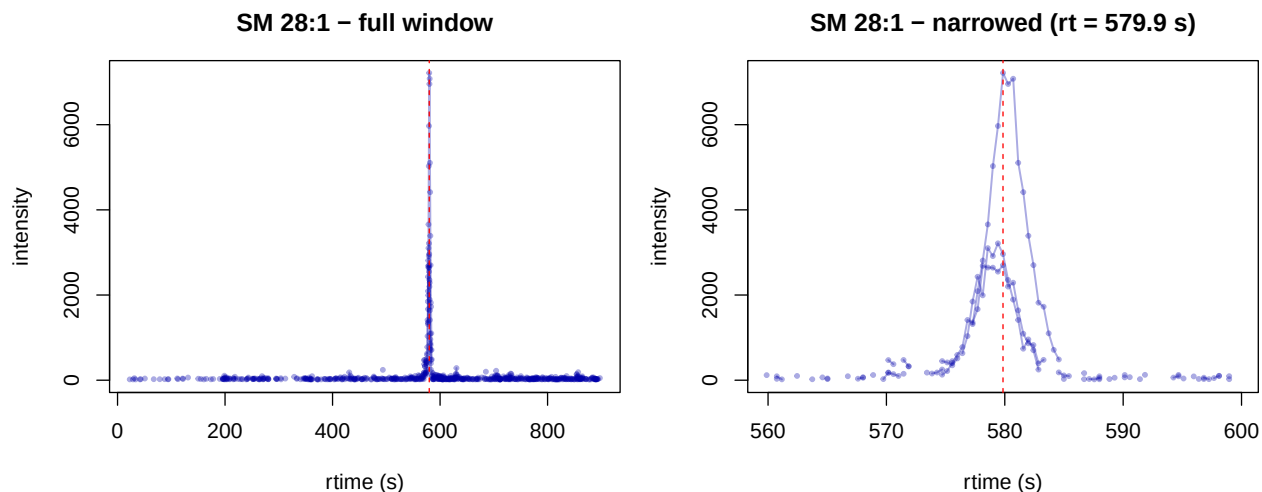


Figure 1: SM 28:1 EIC over the full elution window (left) and narrowed to the located peak (right). Dashed line: consensus apex.

With its retention time fixed at 579.85 s, SM 28:1 joins the other three targets and is treated identically from here on.

3. Extract EICs with *chromExtract()*

The target table is expanded to one row per (target, file). `chromExtract()` requires `mzMin`, `mzMax`, `rtMin`, `rtMax` and the `by` columns. Every other column (here the compound name and expected `m/z` / `rt`) is carried through and preserved on the output `chromData`.

```

peak_table <- data.frame(
  name = targets$name, target_mz = targets$mz, target_rt = targets$rt,
  mzMin = targets$mzmin, mzMax = targets$mzmax,
  rtMin = targets$rtmin, rtMax = targets$rtmax,
  msLevel = 1L, dataOrigin = rep(files, each = nrow(targets)))

chr <- chromExtract(Chromatograms(sps), peak.table = peak_table,
  by = c("msLevel", "dataOrigin"))
chr <- filterEmptyChromatograms(chr)
chr

```

Chromatographic data (Chromatograms) with 12 chromatograms in a ChromBackendSpectra backend:

```

chromIndex msLevel mz

```

```

1      NA      1 Inf
2      NA      1 Inf
3      NA      1 Inf
4      NA      1 Inf
5      NA      1 Inf
6      NA      1 Inf
... 10 more chromatogram variables/columns
... 2 peaksData variables

```

The Spectra object contains 5949 spectra

Processing:

Filter: removed empty chromatograms. [Mon Jul 27 12:27:12 2026]

The external metadata are preserved on chromData:

Table 4: chromData of extracted EICs (first 6 rows)

name	target_mz	target_rt	mzMin	mzMax	rtMin	rtMax	dataOrigin
LysoPC_18:5	22.356	382.20	522.346	522.366	362.20	402.20	MTBLS93_1_021228_TwinGene_000101.CI
LysoPC_18:3	20.341	346.80	520.331	520.351	326.80	366.80	MTBLS93_1_021228_TwinGene_000101.CI
MG_18:2	337.269	384.00	337.259	337.279	364.00	404.00	MTBLS93_1_021228_TwinGene_000101.CI
SM_28:1	661.526	579.85	661.516	661.536	559.85	599.85	MTBLS93_1_021228_TwinGene_000101.CI
LysoPC_18:5	22.356	382.20	522.346	522.366	362.20	402.20	MTBLS93_1_021228_TwinGene_000201.CI
LysoPC_18:3	20.341	346.80	520.331	520.351	326.80	366.80	MTBLS93_1_021228_TwinGene_000201.CI

4. Quantify and quality-assess the EICs

Rather than integrate the traces by hand, we use the package's own peak functions. `peakBoundary()` locates the retention-time boundaries of the tallest peak in each EIC, and *MsQuality*'s `calculateMetrics()` computes abundance and shape metrics (apex intensity, area under the trace, peak width at half maximum). Both are attached to `chromData`.

```

bounds <- peakBoundary(chr)

metrics <- calculateMetrics(
  chr,
  metrics = c("maxIntensity", "areaUnderTic", "xicFwhm"),
  na.rm = TRUE, filterEmptyObject = FALSE, baselineQuantile = 0.1
)
metrics <- as.data.frame(metrics, row.names = NULL)

chromData(chr) <- cbind(chromData(chr),
  left_boundary = bounds[, "left_boundary"],
  right_boundary = bounds[, "right_boundary"],
  metrics)

```

Per-target summary. The LysoPCs are high-abundance and recovered in every sample, MG 18:2 is weaker. The peak boundaries from `peakBoundary()` bracket the annotated retention time in every case.

Table 5: Recovery, quantification and quality per target

name	n_files	with_signal	boundary_brackets_r	median_apex	median_fwhm	target_rt
LysoPC_18:1	3	3	3	462720	4.07	382.20
LysoPC_18:2	3	3	3	580488	3.73	346.80
MG_18:2	3	3	3	1813	3.28	384.00
SM_28:1	3	3	3	3213	3.81	579.85

5. EIC overlay

`plotChromatogramsOverlay()` overlays the extracted EICs across the subset for each target. A clear peak appears near the annotated retention time (dashed line), with the sample-to-sample retention-time drift characteristic of this cohort.

```
cd <- chromData(chr)
par(mfrow = c(1, nrow(targets)), mar = c(4, 4, 3, 1))
for (nm in targets$name) {
  tgt <- targets[targets$name == nm, ]
  idx <- which(cd$name == nm)
  Chromatograms::plotChromatogramsOverlay(
    chr[idx], col = "#0000AA55",
    main = sprintf("%s (m/z %.3f)", nm, tgt$mz))
  abline(v = tgt$rt, col = "grey60", lty = 2)
}
```

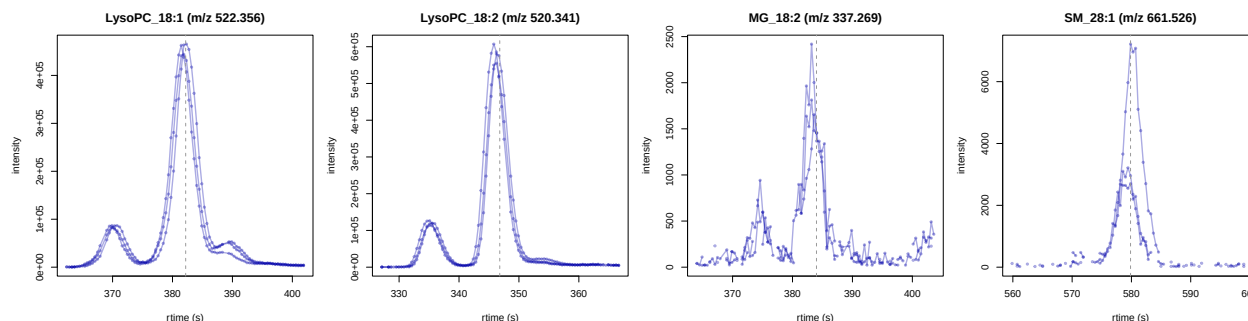


Figure 2: Extracted-ion chromatograms of the annotated biomarkers across the subset (`plotChromatogramsOverlay()`). Dashed line: annotated retention time.

6. Scaling across the cohort

The targeted workflow processes each file independently, so it scales with the number of files. We sweep the file count and record the time and peak RAM to extract all target EICs (materialised into memory with `setBackend()`), using `peakRAM`. This is the throughput of the extraction itself, over the same four biomarkers.

```
file_counts <- seq_len(N_FILES) # test run; the full cohort is swept separately

throughput <- do.call(rbind, lapply(file_counts, function(n) {
  fsub <- files[seq_len(n)]
```

```

ssub <- sps[dataOrigin(sps) %in% fsub]
pt <- peak_table[peak_table$dataOrigin %in% fsub, ]
pr <- peakRAM(setBackend(Chromatograms(ssub, chromData = pt),
                             ChromBackendMemory()))

data.frame(n_files = n,
            extract_s = pr$Elapsed_Time_sec, peak_MB = pr$Peak_RAM_Used_MiB)
)))
knitr::kable(throughput, digits = 2, row.names = FALSE,
              caption = "chromExtract throughput vs number of files")

```

Table 6: chromExtract throughput vs number of files

n_files	extract_s	peak_MB
1	3.27	718.1
2	10.16	812.6
3	37.31	813.5

```

par(mfrow = c(1, 2))
plot(throughput$n_files, throughput$extract_s, type = "b", pch = 19,
     col = "#1B9E77", xlab = "number of files", ylab = "seconds",
     main = "Extraction time")
plot(throughput$n_files, throughput$peak_MB, type = "b", pch = 19,
     col = "#D95F02", xlab = "number of files", ylab = "peak RAM (MiB)",
     main = "Peak memory")

```

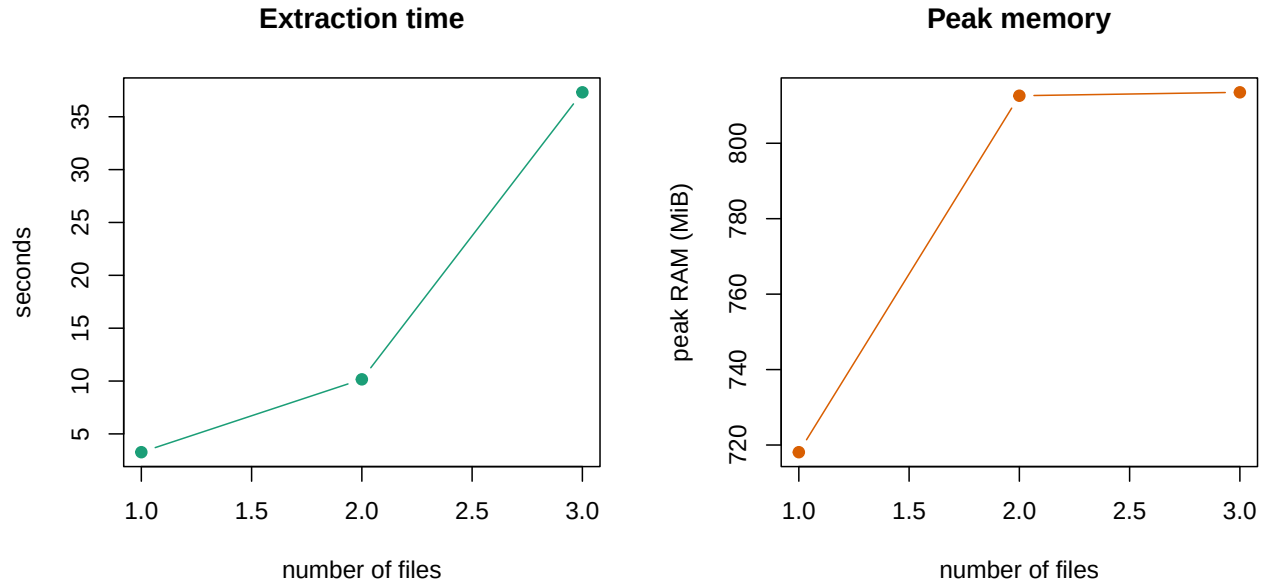


Figure 3: Targeted extraction time and peak RAM vs number of files.

Key takeaways

Starting from a study's published annotations (m/z + retention time) and its raw public files, `chromExtract()` recovers the target compounds across a cohort in a single pass over the data.

The untargeted route on the same cohort (the Metabonaut large-scale-analysis tutorial) requires many hours of *xcms* preprocessing. Targeted reanalysis with `chromExtract()` is a lightweight, annotation-driven alternative.

Session Info

```
sessionInfo()
```

```
R version 4.6.0 (2026-04-24 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)
```

```
Matrix products: default
  LAPACK version 3.12.1
```

```
locale:
[1] LC_COLLATE=English_United Kingdom.utf8
[2] LC_CTYPE=English_United Kingdom.utf8
[3] LC_MONETARY=English_United Kingdom.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United Kingdom.utf8
```

```
time zone: Europe/Paris
tzcode source: internal
```

```
attached base packages:
[1] stats4      stats      graphics  grDevices  utils      datasets  methods
[8] base
```

```
other attached packages:
[1] peakRAM_1.0.2           MsQuality_1.12.0
[3] MsBackendMetaboLights_1.7.4 Spectra_1.22.2
[5] S4Vectors_0.50.1        BiocGenerics_0.58.1
[7] generics_0.1.4           Chromatograms_1.3.2
[9] ProtGenerics_1.44.0      BiocParallel_1.46.0
```

```
loaded via a namespace (and not attached):
[1] DBI_1.3.0                httr2_1.2.3
[3] testthat_3.3.2           rlang_1.2.0
[5] magrittr_2.0.5           shinydashboard_0.7.3
[7] clue_0.3-68              otel_0.2.0
[9] matrixStats_1.5.0        compiler_4.6.0
[11] RSQLite_3.53.3           png_0.1-9
[13] vctrs_0.7.3              reshape2_1.4.5
```

[15]	stringr_1.6.0	pkgconfig_2.0.3
[17]	MetaboCoreUtils_1.20.1	crayon_1.5.3
[19]	fastmap_1.2.0	dbplyr_2.6.0
[21]	XVector_0.52.0	promises_1.5.0
[23]	rmarkdown_2.31	tinytex_0.60
[25]	purrr_1.2.2	bit_4.6.0
[27]	xfun_0.59	MultiAssayExperiment_1.38.0
[29]	MsDataHub_1.12.0	cachem_1.1.0
[31]	jsonlite_2.0.0	progress_1.2.3
[33]	blob_1.3.0	later_1.4.8
[35]	DelayedArray_0.38.2	parallel_4.6.0
[37]	prettyunits_1.2.0	cluster_2.1.8.2
[39]	R6_2.6.1	stringi_1.8.7
[41]	RColorBrewer_1.1-3	brio_1.1.5
[43]	GenomicRanges_1.64.0	Rcpp_1.1.1-1.1
[45]	Seqinfo_1.2.0	SummarizedExperiment_1.42.0
[47]	knitr_1.51	IRanges_2.46.0
[49]	httpuv_1.6.17	Matrix_1.7-5
[51]	igraph_2.3.3	tidyselect_1.2.1
[53]	abind_1.4-8	yaml_2.3.12
[55]	codetools_0.2-20	curl_7.1.0
[57]	lattice_0.22-9	tibble_3.3.1
[59]	plyr_1.8.9	withr_3.0.3
[61]	shiny_1.14.0	Biobase_2.72.0
[63]	KEGGREST_1.52.2	S7_0.2.2
[65]	evaluate_1.0.5	ontologyIndex_2.12
[67]	BiocFileCache_3.2.0	ExperimentHub_3.2.0
[69]	Biostrings_2.80.1	pillar_1.11.1
[71]	BiocManager_1.30.27	filelock_1.0.3
[73]	MatrixGenerics_1.24.0	ncdf4_1.24
[75]	plotly_4.12.0	BiocVersion_3.23.1
[77]	hms_1.1.4	ggplot2_4.0.3
[79]	scales_1.4.0	R6P_0.4.0
[81]	MsExperiment_1.14.0	xtable_1.8-8
[83]	glue_1.8.1	lazyeval_0.2.3
[85]	tools_4.6.0	AnnotationHub_4.2.1
[87]	data.table_1.18.4	mzR_2.46.0
[89]	QFeatures_1.22.0	fs_2.1.0
[91]	grid_4.6.0	jsonvalidate_1.5.0
[93]	tidyr_1.3.2	MsCoreUtils_1.24.0
[95]	AnnotationDbi_1.74.0	cli_3.6.6
[97]	rappdirs_0.3.4	S4Arrays_1.12.0
[99]	viridisLite_0.4.3	dplyr_1.2.1
[101]	AnnotationFilter_1.36.0	gtable_0.3.6
[103]	digest_0.6.39	SparseArray_1.12.2
[105]	htmlwidgets_1.6.4	farver_2.1.2
[107]	memoise_2.0.1	htmltools_0.5.9
[109]	lifecycle_1.0.5	httr_1.4.8

[111] mime_0.13
[113] bit64_4.8.2

rmzqc_0.7.0
MASS_7.3-65